

2nd Derivative Test cont.

section 3.4

9/29/25

Using 2nd Deriv to find local extrema

$$f'(x) = 0 \text{ or DNE}$$

$$x = a$$

$$f''(a) > 0 \quad \text{local min}$$

$$f''(a) < 0 \quad \text{local max}$$

$$f''(a) = 0 \quad \text{FAIL!} \quad \text{Inconclusive}$$

↳ gotta use first derivative test

$$\text{Ex 1: } f(x) = x^3 - 12x - 5$$

$$f'(x) = 3x^2 - 12 = 0$$

$$x^2 - 4 = 0$$

$$x = \pm 2$$

$$f'(x) = 6x$$

$$f''(2) = 12 > 0$$

↳ f has a local min at $x = 2$
b/c $f'(2) = 0$, and $f''(2) > 0$

$$f''(-2) = -12 < 0$$

↳ f has a local max at $x = -2$
b/c $f'(-2) = 0$ and $f''(-2) < 0$

Ex 2:

$$f'(x) = 3x^2 = 0$$

$$x = 0$$

$$f''(x) = 6x$$

$$f''(0) = 0$$

→ Fails



1st Deriv



f' doesn't change signs, so
ne